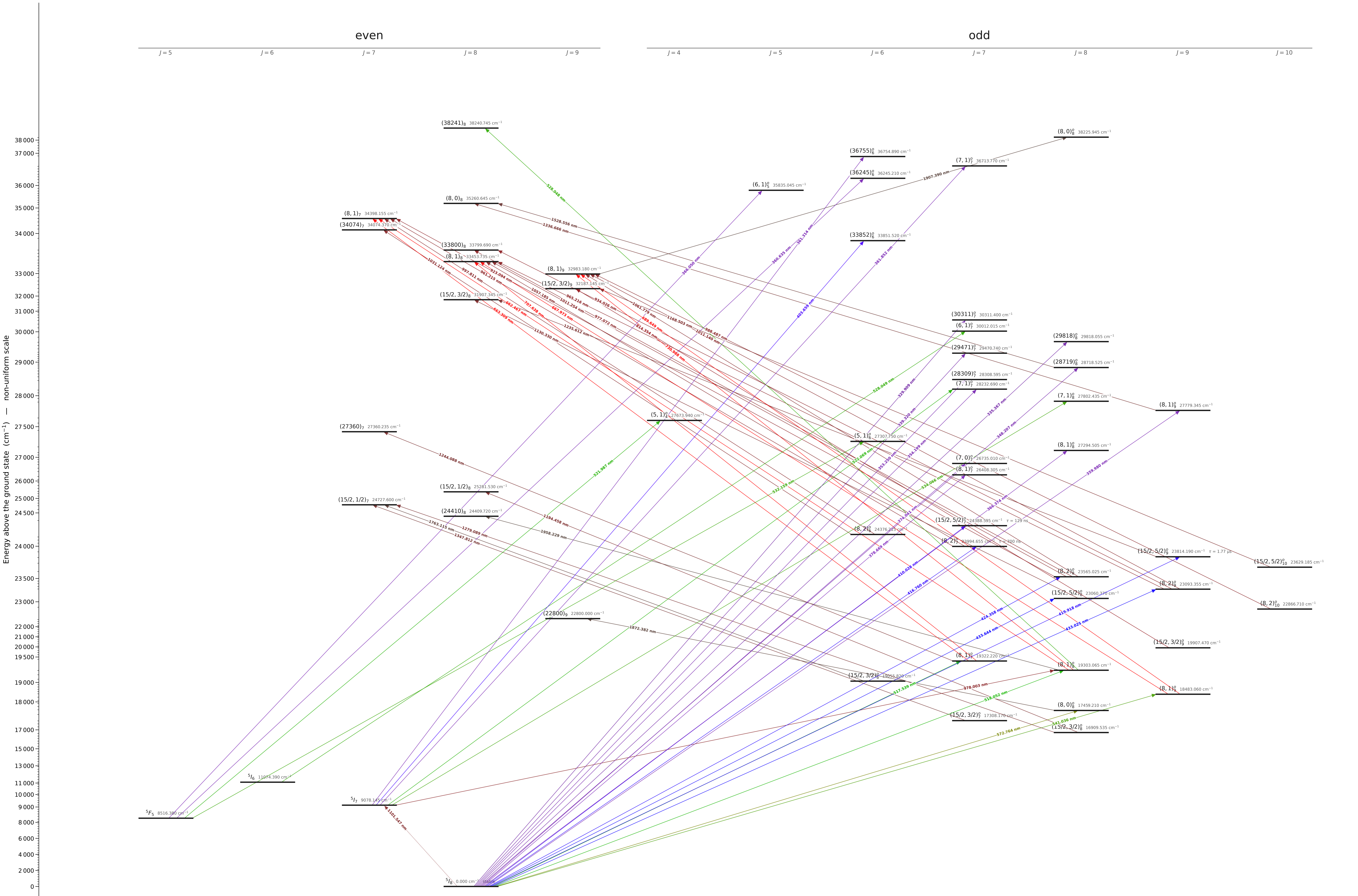
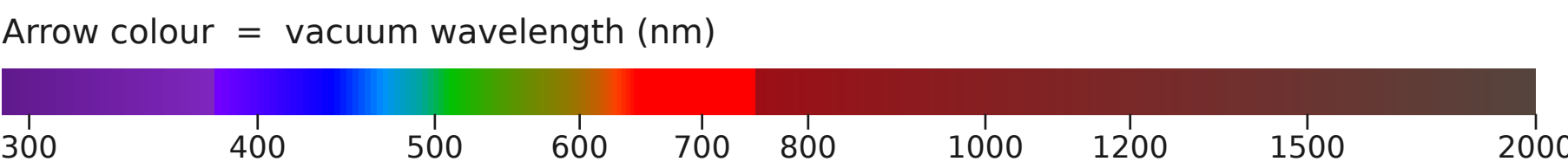


Californium

Cf (Cf I). Classified transitions below 2 μm (levels and lines from the literature): 55 levels, 65 lines.

Cf+ ionisation limit 50666.76 cm⁻¹ = 6.28188 eV (197.368 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U||er||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[⟨J er J'⟩] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ f ₈ – (8, 1) _g	359.9797	359.8770	832.8038	–	–	–	–	– first step of Mainz resonance ionization scheme F
⁵ f ₈ – (8, 1) _g	366.3741	366.2698	818.2687	–	–	–	–	– first step of Mainz resonance ionization scheme E
⁵ f ₈ – (8, 1) _g	378.6688	378.5612	791.7011	–	–	–	–	– first step of Mainz resonance ionization scheme D
⁵ f ₈ – (15/2, 5/2) _g	410.0277	409.9120	731.1517	–	–	1.234	–	– first step of Mainz scheme C / III
⁵ f ₈ – (8, 2) _g	416.7595	416.6420	719.3417	–	0.2274	–	–	– first step of Mainz scheme B / II
⁵ f ₈ – (15/2, 5/2) _g	419.9177	419.7994	713.9315	–	–	0.08992	–	– first step of Mainz two-step resonance ionization scheme...

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁵³Cf hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁵ f ₈	-127(3)	2640(200)	F=9/2: +4941.00 F=11/2: +3749.86 F=13/2: +2464.71 F=15/2: +1146.86 F=17/2: -133.00 F=19/2: -1294.71 F=21/2: -2248.71 F=23/2: -2896.00
(15/2, 5/2) _g	-109(3)	11100(850)	F=11/2: +7623.82 F=13/2: +4793.27 F=15/2: +2071.35 F=17/2: -308.72 F=19/2: -2082.66 F=21/2: -2955.11 F=23/2: -2599.57 F=25/2: -658.50
(8, 2) _g	-28.6(7)	-1130(90)	F=7/2: +378.60 F=9/2: +453.46 F=11/2: +496.17 F=13/2: +477.47 F=15/2: +362.75 F=17/2: +112.11 F=19/2: -319.69 F=21/2: -983.20
(15/2, 5/2) _g	-170(4)	3430(260)	F=7/2: +6041.54 F=9/2: +4658.65 F=11/2: +3116.54 F=13/2: +1504.04 F=15/2: -73.85 F=17/2: -1495.96 F=19/2: -2625.00 F=21/2: -3307.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁵ f ₈ – (15/2, 5/2) _g	410.028	249-250	-5921(12)
⁵ f ₈ – (15/2, 5/2) _g	410.028	251-250	5013(12)
⁵ f ₈ – (15/2, 5/2) _g	410.028	252-250	11265(14)
⁵ f ₈ – (15/2, 5/2) _g	410.028	253-250	15230(75)
⁵ f ₈ – (8, 2) _g	416.760	249-250	6250(12)
⁵ f ₈ – (8, 2) _g	416.760	251-250	-5274(12)
⁵ f ₈ – (8, 2) _g	416.760	252-250	-11887(14)
⁵ f ₈ – (8, 2) _g	416.760	253-250	-16052(20)
⁵ f ₈ – (15/2, 5/2) _g	419.918	249-250	-6812(12)
⁵ f ₈ – (15/2, 5/2) _g	419.918	251-250	5721(14)
⁵ f ₈ – (15/2, 5/2) _g	419.918	252-250	12944(14)
⁵ f ₈ – (15/2, 5/2) _g	419.918	253-250	17470(50)

Shift = ν(first isotope) – ν(second isotope).